

Erin Weiss

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Portfolio: erin-weiss.github.io

SUMMARY

Data Scientist and ML Engineer with experience building and deploying end-to-end machine learning systems. Trained gradient-boosted and deep learning models on datasets of 200k+ records and deployed them as production APIs with containerized infrastructure, monitoring, and CI/CD. Background in strategic planning and business analytics provides a unique ability to connect model outputs to measurable business outcomes. Known for exceptional organization, analytical problem-solving, and communicating complex technical concepts to diverse audiences.

SKILLS

ML & Modeling: Machine Learning, Deep Learning, MLOps, CatBoost, scikit-learn, TensorFlow, Keras, NLP, Computer Vision, CNNs, Reinforcement Learning, Recommender Systems, Time Series Analysis, Predictive Modeling, Feature Engineering, PCA, Optimization, Gurobi

MLOps & Deployment: FastAPI, Docker, Kubernetes, GitHub Actions (CI/CD), Prometheus, Grafana, Pydantic, Uvicorn, pytest

Data & Analysis: Python, R, SQL, pandas, NumPy, PySpark, Apache Spark, Matplotlib, Seaborn, Jupyter, Quarto, Statistical Modeling, Data Visualization

Databases & Data Engineering: PostgreSQL, MySQL, Microsoft SQL Server, SQLite, Hadoop, Google Cloud Dataproc, Data Warehousing, ETL, Alteryx, SSRS

Tools & Platforms: Git, GitHub, Bash, Streamlit, Tableau, Google Analytics, LaTeX, Figma, HTML, CSS, Microsoft Office, Excel

PROJECTS

Used Car Price Prediction — Model Development | [GitHub Repo](#)

- Trained a CatBoost model predicting used-car prices within ~\$1,300 at the median across a 243k-record dataset spanning 29 manufacturers and 5,600+ model variants
- Compared three architectures (Ridge regression, CatBoost, FT-Transformer) and selected CatBoost for production based on accuracy, inference speed, and native categorical feature handling
- Engineered 20+ features from raw vehicle attributes; managed reproducible experiments with versioned run artifacts

Used Car Price Prediction API — Production Deployment | [GitHub Repo](#)

- Deployed the trained model as a production API using FastAPI, Docker, and Kubernetes with HPA autoscaling, Prometheus + Grafana monitoring, and CI/CD via GitHub Actions
- Implemented request validation with Pydantic v2, fuzzy input matching for free-text fields, and median imputation for optional features
- Built a full test suite including unit tests, integration tests, and end-to-end predictions against the real model, running automatically on every merge to main

EXPERIENCE

Innovative Data Solutions, Remote

Aug 2024 – Present

Independent Data Scientist

- Built and deployed a Streamlit web application for a client to clean, standardize, and reconcile Excel invoice data, reducing manual processing time
- Developed a Python pipeline to automate recurring data exports into formatted Excel reports with conditional formatting, threshold analysis, and embedded visualizations

College of William & Mary, Remote

July 2023 – Aug 2024

Python Tutor

- Created tutorials for graduate students in the OMSBA Python course; simplified complex subjects into digestible material
- Collaborated with the professor to integrate student feedback and improve future course content

- Recruited to tutor immediately after completing the course due to outstanding performance; received consistent positive feedback from students

Spirits & Spice, Remote

Feb 2020 – Nov 2022

Director of Strategic Planning

- Gathered and analyzed data to forecast and plan key strategies for the company including shipping costs, cost of goods, sales, marketing, wages, and rent
- Designed and implemented strategies to drive traffic and sales, leading to 2x year-over-year growth on the website in the first year
- Managed marketing strategies end-to-end: allocated budgets, approved initiatives, and kept the team on timeline
- Negotiated contracts for marketing, retail leasing, and contractors

Global Business Policy Council (GBPC) — Kearney, Rosslyn, Virginia

May 2018 – Feb 2020

Communications Coordinator

- Created and managed timelines, deliverables, milestones, and communications strategy within the GBPC team
- Collaborated with the firm's web team to update the GBPC corporate website, translating thought-leadership material into a web-relevant format

EDUCATION

William & Mary, Raymond A. Mason School of Business

Aug 2024

Master of Science in Business Analytics

Georgetown University, School of Foreign Service

May 2018

Bachelor of Science, Science, Technology, and International Affairs (STIA); Minor — Chinese

ADDITIONAL INFORMATION

Leadership: Managed teams, projects, and individuals in roles as Director of Strategic Planning and Communications Coordinator

Languages: Fluent in English and proficient in Mandarin. Passed Georgetown SFS Fluency Exam (Spring 2018); Attained HSK Certification Level II (Fall 2014) after self-teaching Chinese

Volunteer: After School Kids (A.S.K.) Volunteer, Smithsonian Volunteer